



MODULE 4

Influence Lines of Determinate Structures

STRUCTURAL THEORY

Course Description: A professional course common to all civil engineering students designated to provide fundamental concepts, principles, and theories in the theory of structures and structural analysis for internal actions in structure and its deformations under load

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MODULE 4

Influence Lines for Statically Determinate Structures

Overview	
Influence lines have important application for the design of structures that resist large live loads. In this module we will discuss how to draw the influence line for a statically determinate structure. The theory is applied to structures subjected to a distributed load or a series of concentrated forces, and specific applications to floor girders and bridge trusses are given. The determination of the absolute maximum live shear and moment in a member is discussed at the end of this module	
Learning Outcomes:	
APPLY the methods of structural analysis for external and internal actions on various types of structural systems due to static and moving loads.	
Learning Objectives	
Student Learning Objectives for this Module	Upon completion of this module, you will be able to: - Know the meaning of influence lines. - Determine its an important part in the design of bridges, industrial crane rails, conveyors, and other structures where loads move across their span.
Learning Activities	
Activities for this module	- Do the assignments and problem sets - Take the quiz (Quiz1)
Graded Assessment	
Evidence to proceed	Completed all the assessment tasks in this module
References	
Printed Sources	- National Structural Code of the Philippines, 5th ed. Association of Structural Engineers of the Philippines, Manila 2001 by Association of Structural Engineers of the Philippines - Structural Analysis, 4th Ed, Prentice Hall 1999., N.J. By: Hibbler, R.C. - Structural Analysis, By: Besavilla - Structural Analysis, By: Gillesania

Influence Lines

If a structure is subjected to a live or moving load, however, the variation of the shear and bending moment in the member is best described using

the influence line. An influence line represents the variation of the reaction, shear, moment, or deflection at a specific point in a member as a concentrated force moves over the member. Once this line is constructed, one can tell at a glance where the moving load should be placed on the structure so that it creates the greatest influence at the specified point. Furthermore, the magnitude of the associated reaction, shear, moment, or deflection at the point can then be calculated from the ordinates of the influence-line diagram. For these reasons, influence lines play an important part in the design of bridges, industrial crane rails, conveyors, and other structures where loads move across their span.

Procedure for Analysis

Either of the following two procedures can be used to construct the influence line at a specific point P in a member for any function (reaction, shear, or moment). For both of these procedures we will choose the moving force to have a *dimensionless magnitude of unity*.*

Tabulate Values

- Place a unit load at various locations, x , along the member, and at *each* location use statics to determine the value of the function (reaction, shear, or moment) at the specified point.
- If the influence line for a vertical force *reaction* at a point on a beam is to be constructed, consider the reaction to be *positive* at the point when it acts *upward* on the beam.
- If a shear or moment influence line is to be drawn for a point, take the shear or moment at the point as positive according to the same sign convention used for drawing shear and moment diagrams.
- All statically determinate beams will have influence lines that consist of straight line segments. After some practice one should be able to minimize computations and locate the unit load *only* at points representing the *end points* of each line segment.
- To avoid errors, it is recommended that one first construct a table, listing "unit load at x " versus the corresponding value of the function calculated at the specific point; that is, "reaction R ," "shear V ," or "moment M ." Once the load has been placed at various points along the span of the member, the tabulated values can be plotted and the influence-line segments constructed.

Influence-Line Equations

- The influence line can also be constructed by placing the unit load at a *variable*

position x on the member and then computing the value of R , V , or M at the point as a function of x . In this manner, the equations of the various line segments composing the influence line can be determined and plotted.

Influence Lines for Beams

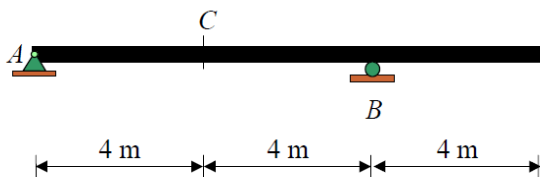
Since beams (or girders) often form the main load-carrying elements of a floor system or bridge deck, it is important to be able to construct the influence lines for the reactions, shear, or moment at any specified point in a beam.

Loadings. Once the influence line for a function (reaction, shear, or moment) has been constructed, it will then be possible to position the live loads on the beam which will produce the maximum value of the function. Two types of loadings will now be considered.

Example 1

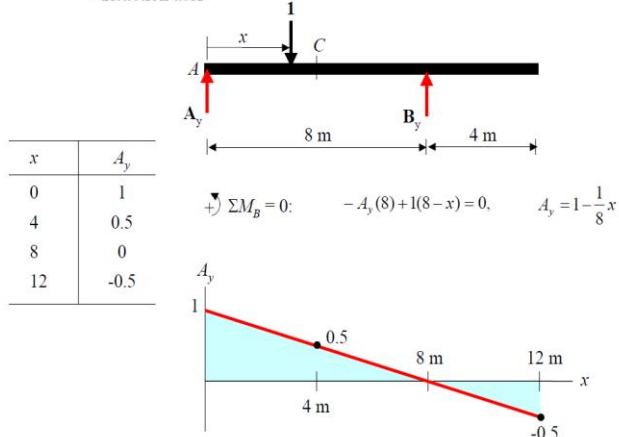
Construct the influence line for

- reaction at A and B
- shear at point C
- bending moment at point C
- shear before and after support B
- moment at point B of the beam in the figure below.

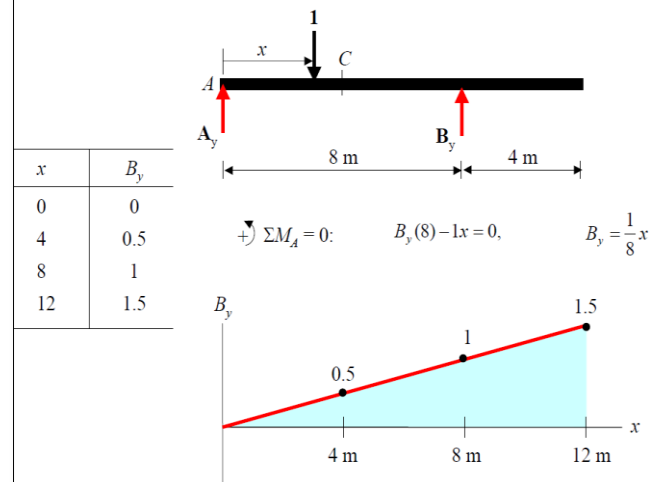


SOLUTION

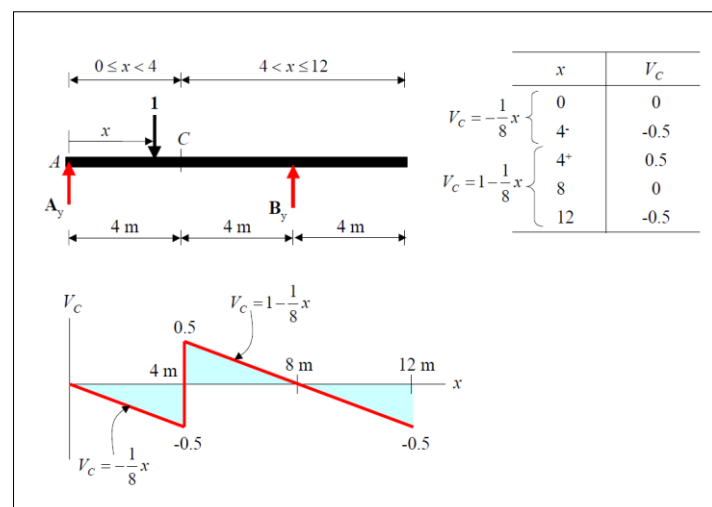
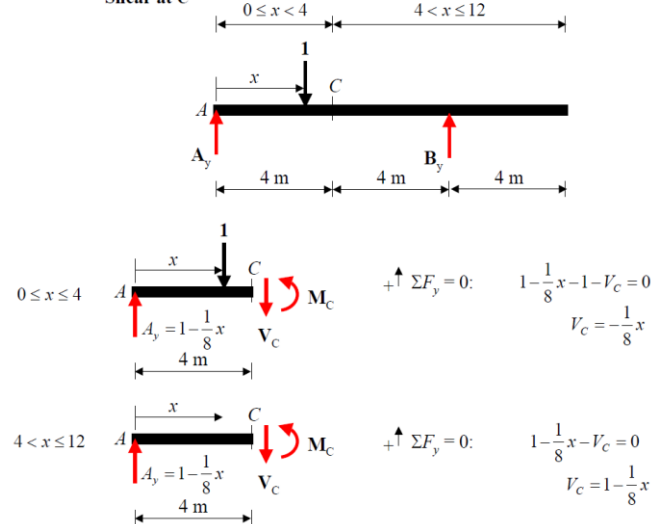
• Reaction at A



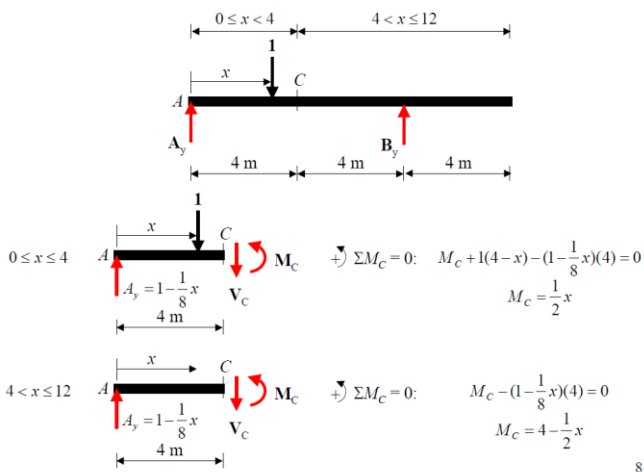
• Reaction at B



• Shear at C

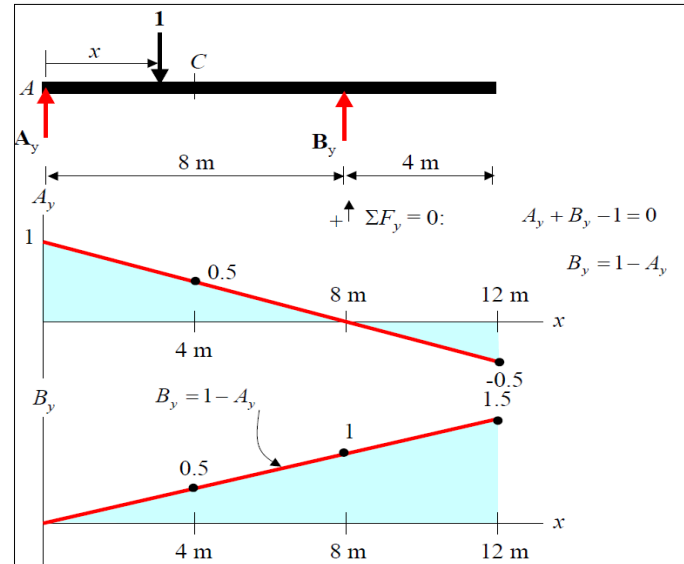


• Bending moment at C

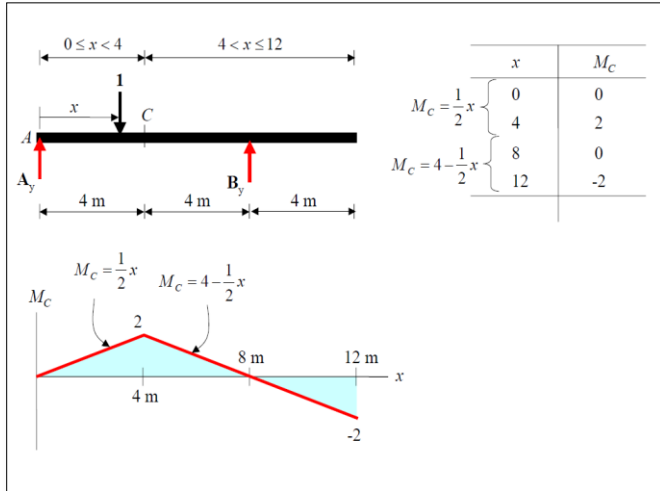


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Reaction at B

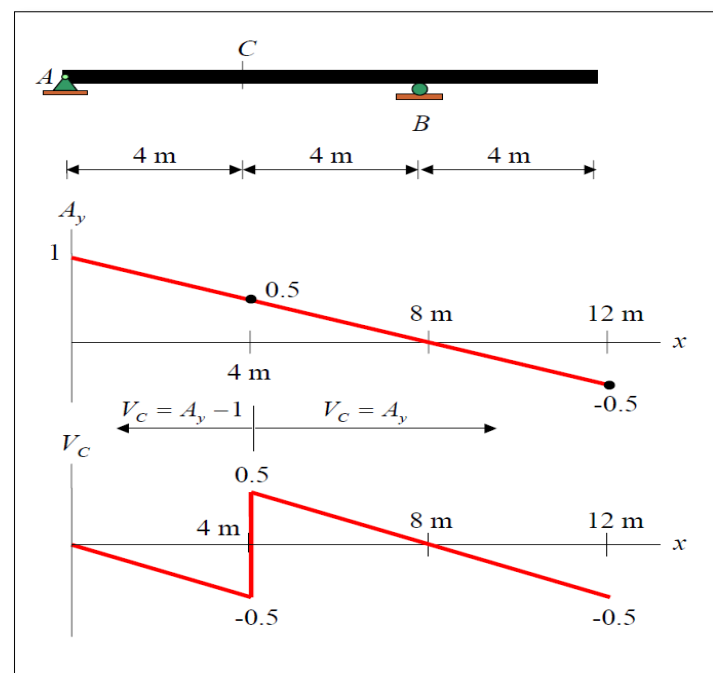
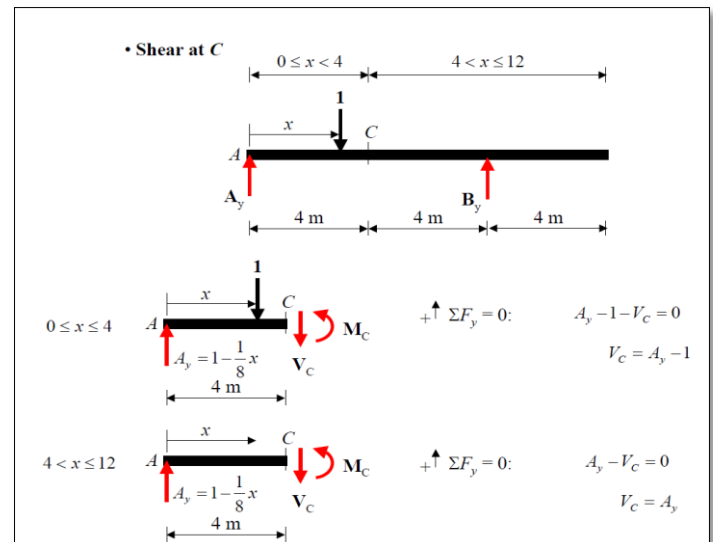
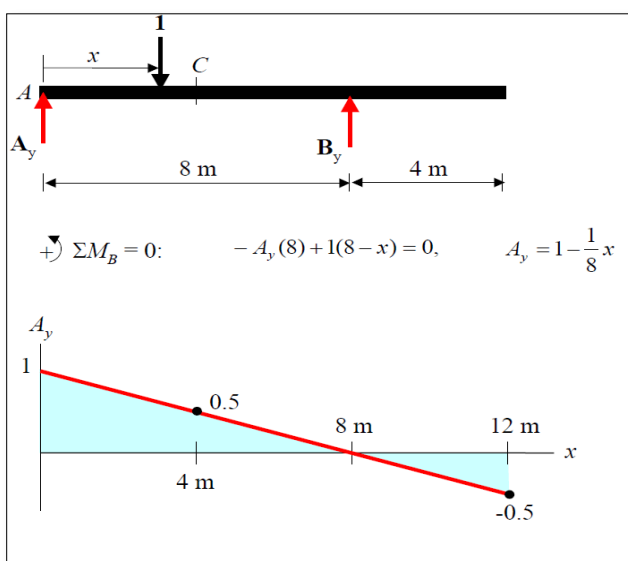


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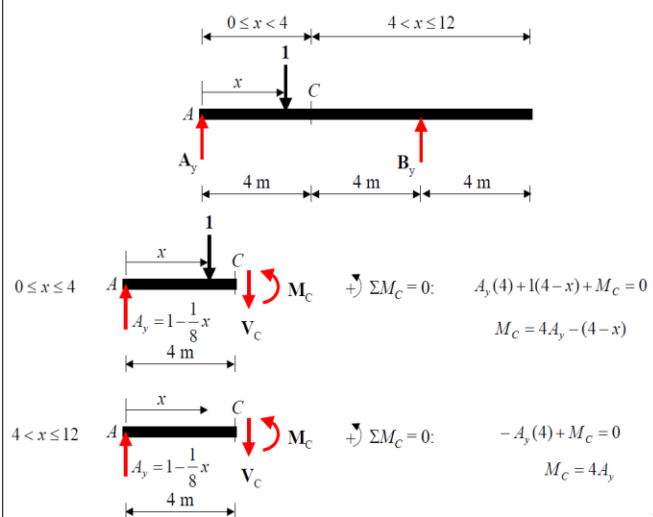


Or Using Equilibrium Conditions

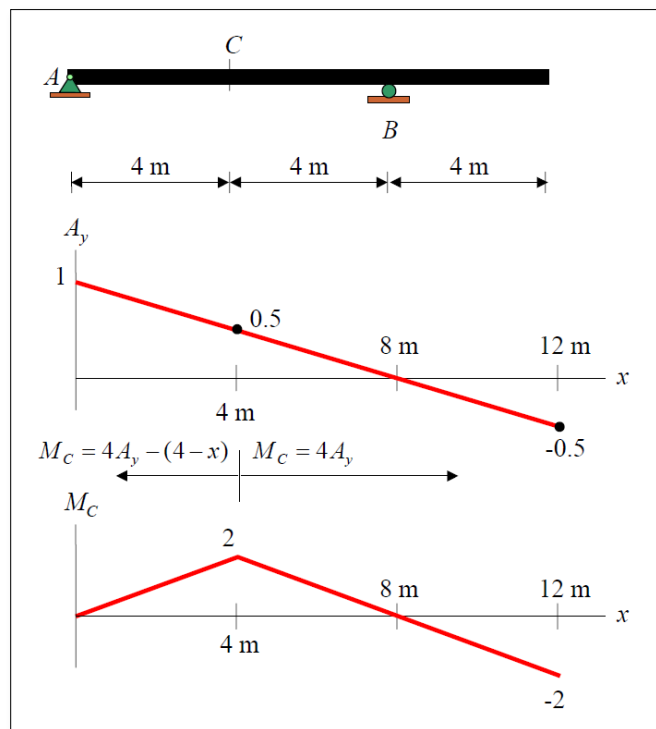
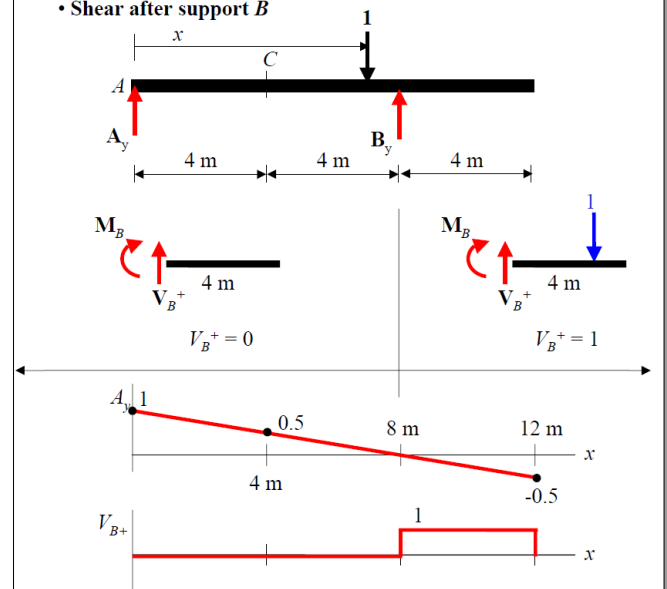
Reaction at A



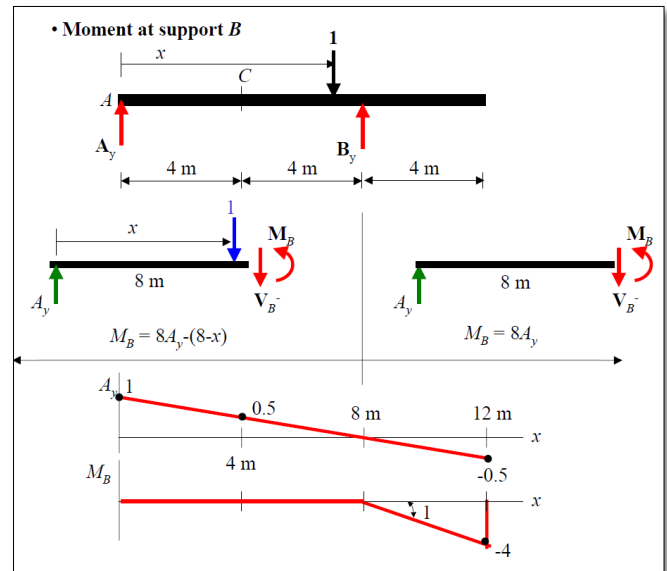
• Bending moment at C



• Shear after support B

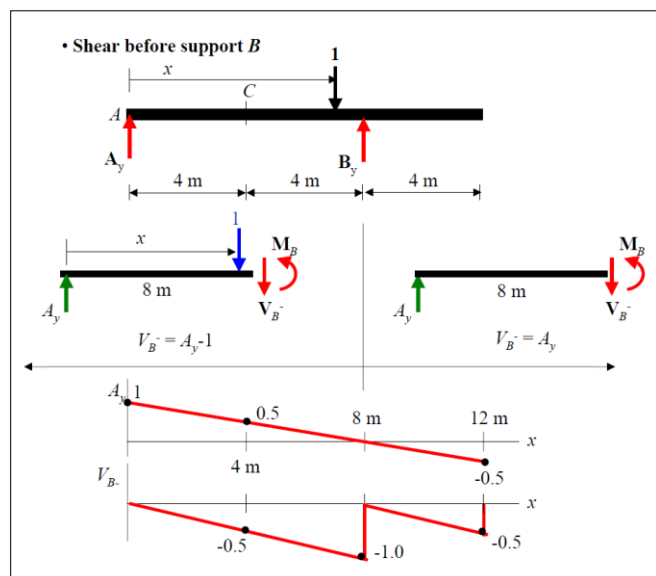
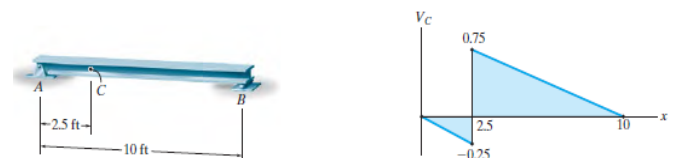


• Moment at support B



Example 2

Determine the maximum positive shear that can be developed at point C in the beam shown in figure due to a concentrated moving load of 4000 lb and a uniform moving load of 2000 lb/ft.



Concentrated Force. The maximum positive shear at C will occur when the 4000-lb force is located at $x = 2.5^+$ ft, since this is the positive peak of the influence line. The ordinate of this peak is $+0.75$; so that

$$V_C = 0.75(4000 \text{ lb}) = 3000 \text{ lb}$$

Uniform Load. The uniform moving load creates the maximum positive influence for V_C when the load acts on the beam between $x = 2.5^+$ ft and $x = 10$ ft, since within this region the influence line has a positive area. The magnitude of V_C due to this loading is

$$V_C = \left[\frac{1}{2}(10 \text{ ft} - 2.5 \text{ ft})(0.75) \right] 2000 \text{ lb/ft} = 5625 \text{ lb}$$

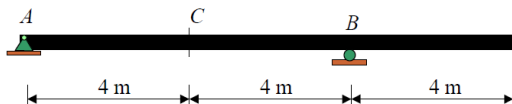
Total Maximum Shear at C.

$$(V_C)_{\max} = 3000 \text{ lb} + 5625 \text{ lb} = 8625 \text{ lb} \quad \text{Ans.}$$

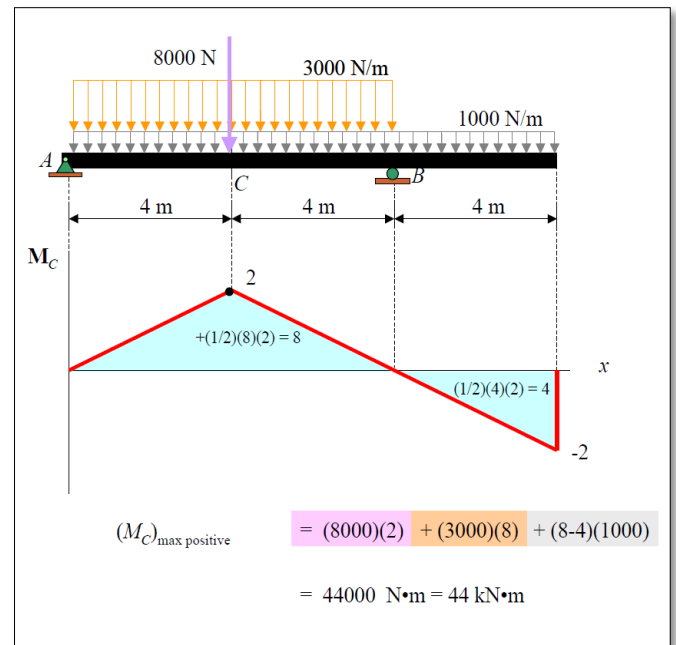
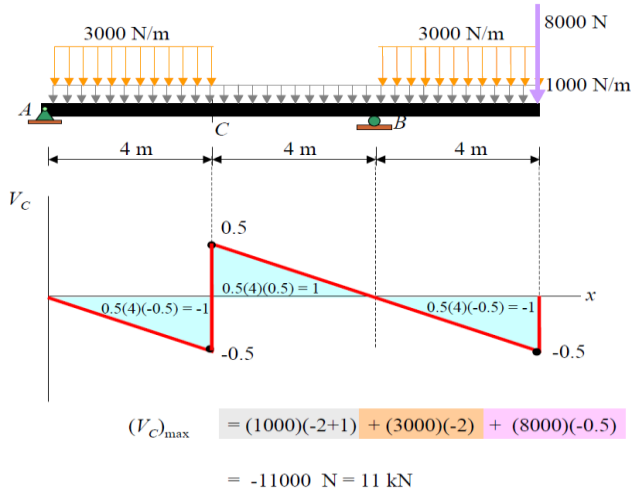
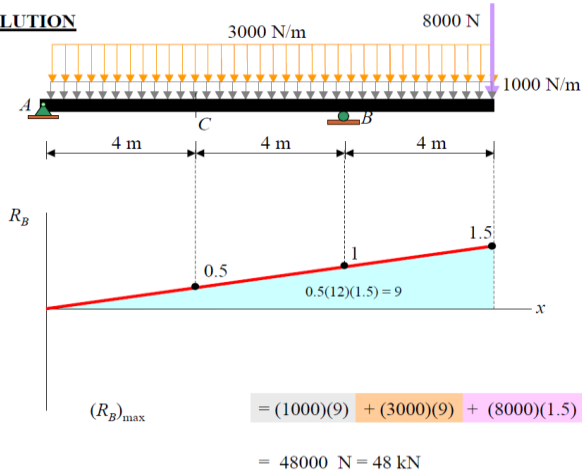
Example 3

Determine the maximum reaction at support B , the maximum shear at point C and the maximum **positive** moment that can be developed at point C on the beam shown due to

- a single concentrate live load of 8000 N
- a uniform live load of 3000 N/m
- a beam weight (dead load) of 1000 N/m



SOLUTION

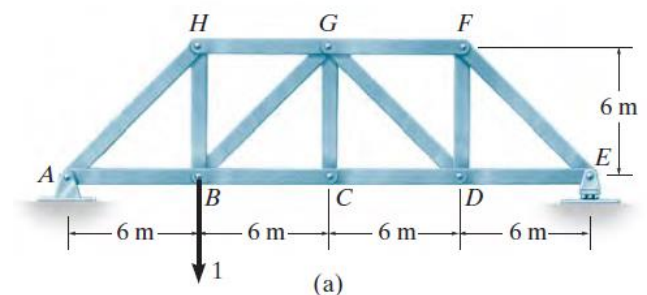


Influence Lines for Trusses

Trusses are often used as primary load-carrying elements for bridges. Hence, for design it is important to be able to construct the influence lines for each of its members. Since the truss members are affected only by the joint loading, we can therefore obtain the ordinate values of the influence line for a member by loading each joint along the deck with a unit load and then use the method of joints or the method of sections to calculate the force in the member. The data can be arranged in tabular form, listing "unit load at joint" versus "force in member." As a convention, if the member force is *tensile* it is considered a *positive* value; if it is *compressive* it is *negative*. The influence line for the member is constructed by plotting the data and drawing straight lines between the points.

Example 1

Draw the influence line for the force in member GB of the bridge truss shown in figure.



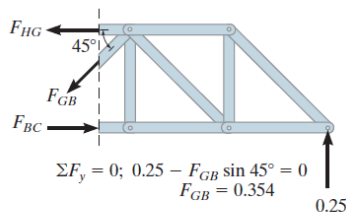
SOLUTION

Tabulate Values. Here each successive joint at the bottom cord is loaded with a unit load and the force in member GB is calculated using the method of sections, Fig. 6–24b. For example, placing the unit load at $x = 6$ m (joint B), the support reaction at E is calculated first, Fig. 6–24a, then passing a section through HG , GB , BC and isolating the right segment, the force in GB is determined, Fig. 6–24c. In the same manner, determine the other values listed in the table.

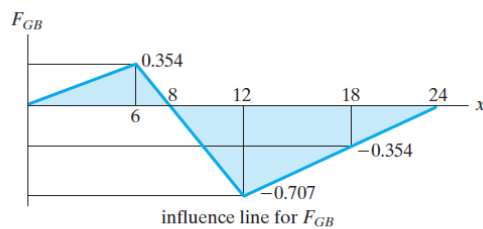
x	F_{GB}
0	0
6	0.354
12	-0.707
18	-0.354
24	0

(b)

Influence Line. Plotting the tabular data and connecting the points yields the influence line for member GB , Fig. 6–24d. Since the influence line extends over the entire span of the truss, member GB is referred to as a *primary member*. This means GB is subjected to a force regardless of where the bridge deck (roadway) is loaded, except, of course, at $x = 8$ m. The point of zero force, $x = 8$ m, is determined by similar triangles between $x = 6$ m and $x = 12$ m, that is, $(0.354 + 0.707)/(12 - 6) = 0.354/x'$, $x' = 2$ m, so $x = 6 + 2 = 8$ m.



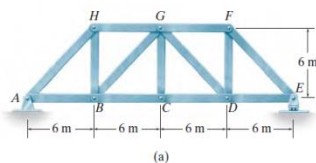
(c)



(d)

Example 2

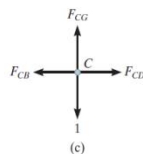
Draw the influence line for the force in member CG of the bridge truss shown in figure.



(a)

x	F_{CG}
0	0
6	0
12	1
18	0
24	0

(b)

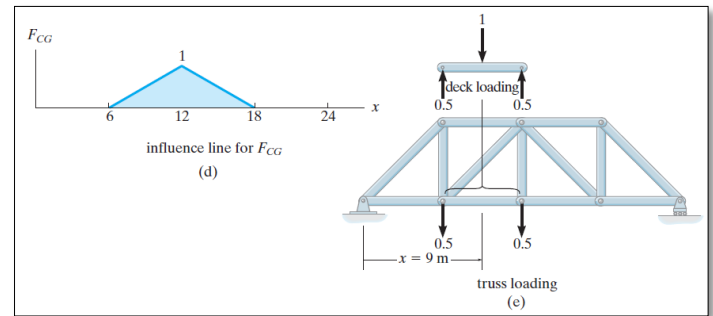


(c)

SOLUTION

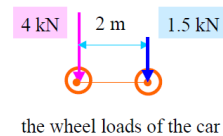
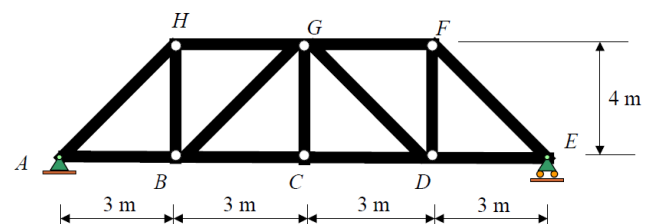
Tabulate Values. A table of unit-load position at the joints of the bottom cord versus the force in member CG is shown in Fig. 6–25b. These values are easily obtained by isolating joint C , Fig. 6–25c. Here it is seen that CG is a zero-force member unless the unit load is applied at joint C , in which case $F_{CG} = 1$ (T).

Influence Line. Plotting the tabular data and connecting the points yields the influence line for member CG as shown in Fig. 6–25d. In particular, notice that when the unit load is at $x = 9$ m, the force in member CG is $F_{CG} = 0.5$. This situation requires the unit load to be placed on the bridge deck *between* the joints. The transference of this load from the deck to the truss is shown in Fig. 6–25e. From this one can see that indeed $F_{CG} = 0.5$ by analyzing the equilibrium of joint C , Fig. 6–25f. Since the influence line for CG does *not* extend over the entire span of the truss, Fig. 6–25d, member CG is referred to as a *secondary member*.



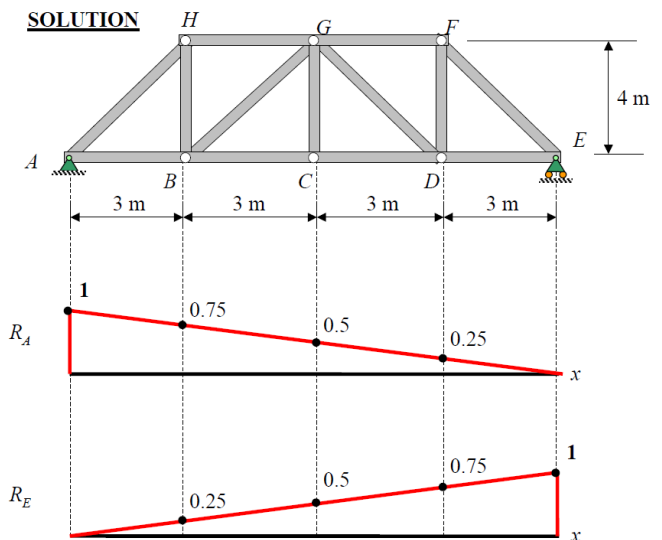
Example 3

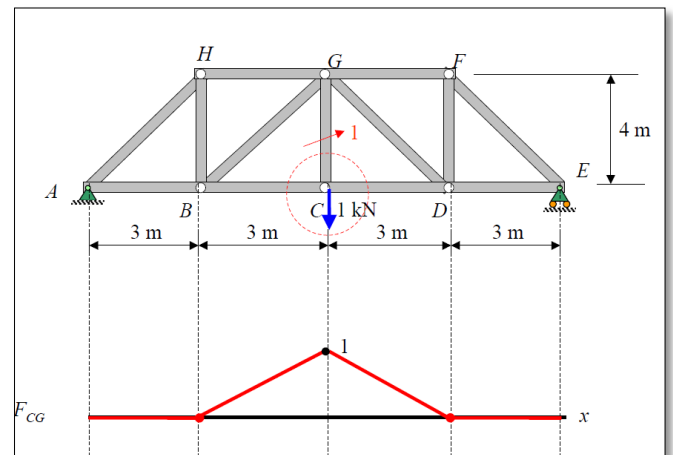
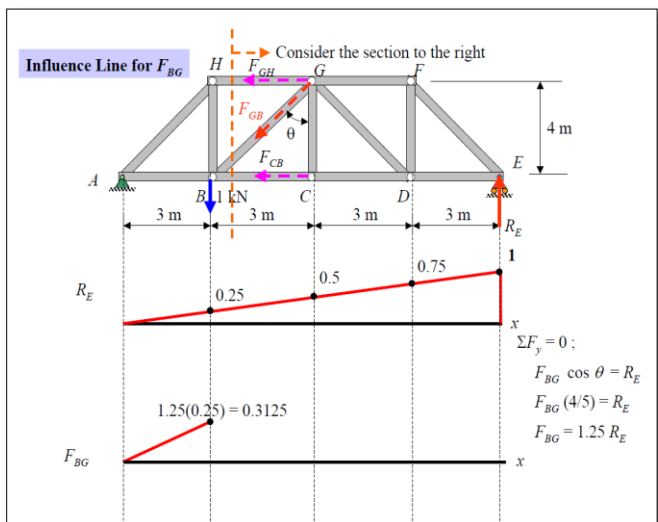
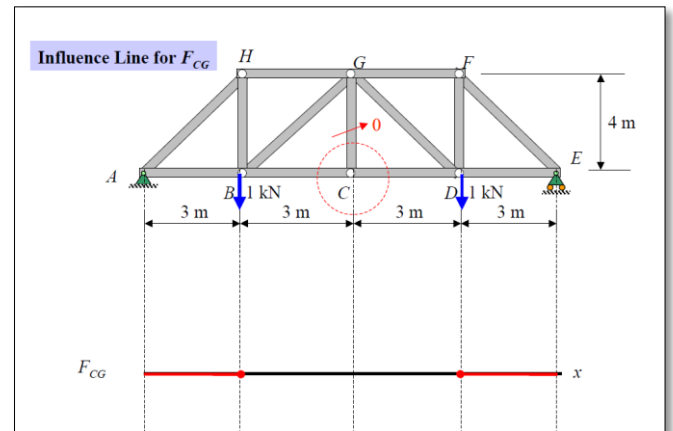
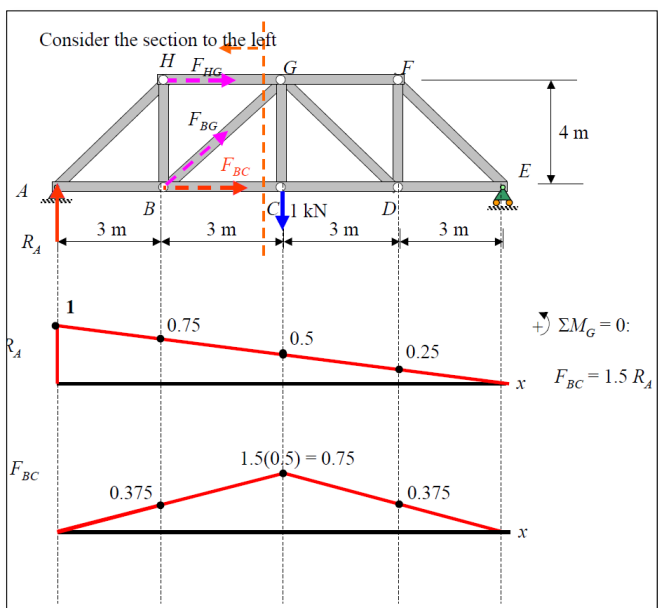
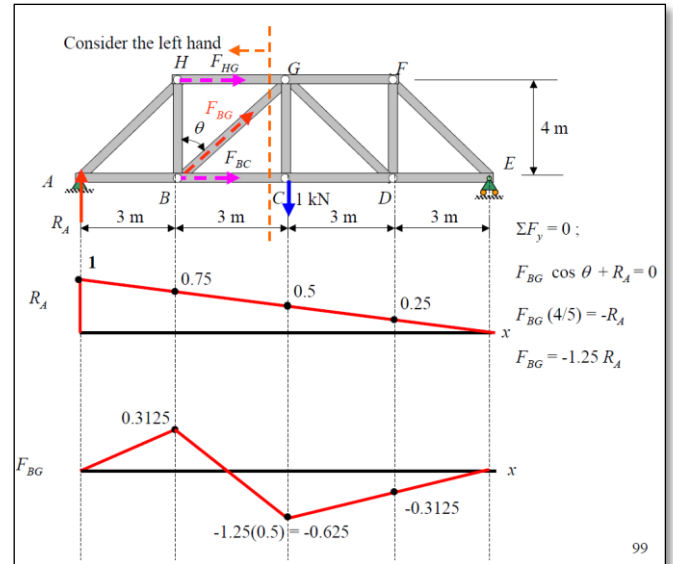
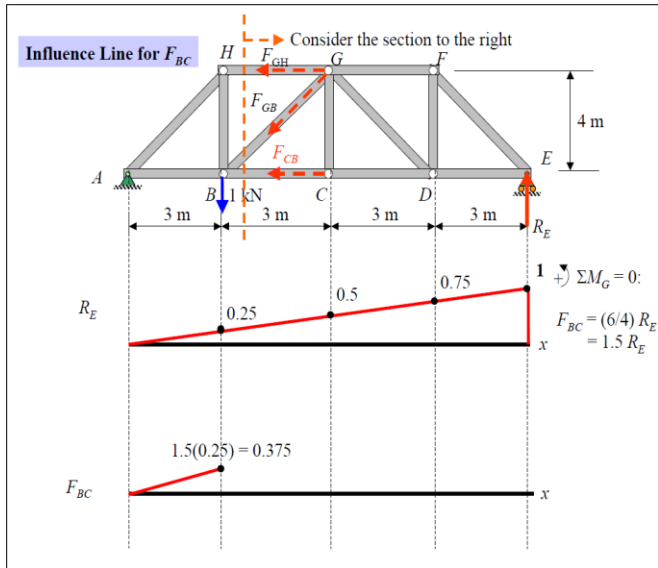
Determine the maximum force developed in member BC , BG , and CG of the truss due to the wheel loads of the car. Assume the loads are applied directly to the truss.



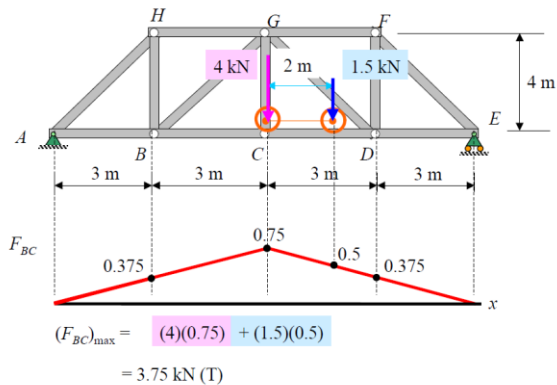
the wheel loads of the car

SOLUTION

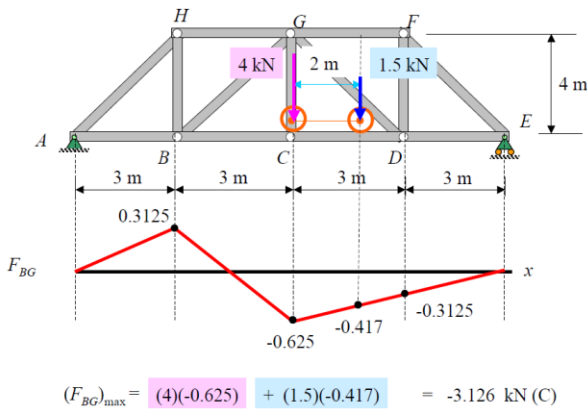




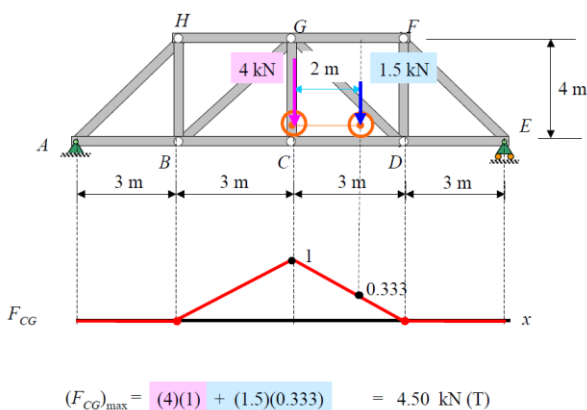
$(F_{BC})_{\max}$ by Loads



$(F_{BG})_{\max}$ by Loads



$(F_{CG})_{\max}$ by Loads



Maximum Influence at a Point due to a Series of Concentrated Loads

Once the influence line of a function has been established for a point in a structure, the maximum effect caused by a live concentrated force is determined by multiplying the peak ordinate of the influence line by the magnitude of the force. In some cases, however, several concentrated forces must be placed on the structure. An example would be the wheel loadings of a truck or train. In order to determine the maximum effect in this case, either a trial-and-error procedure can be used or a method that is based on the change in the function that takes place as the load is moved. Each of these methods will now be explained specifically as it applies to shear and moment.

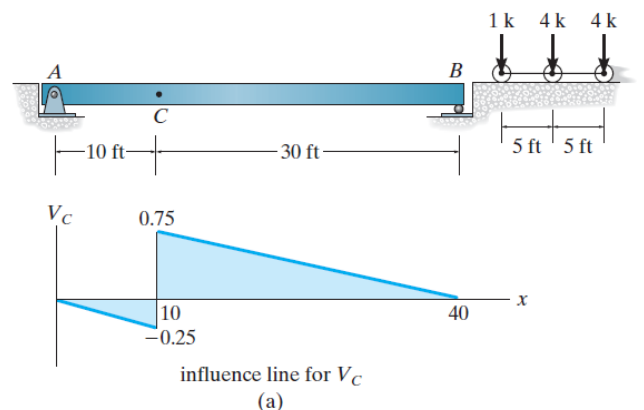
Shear. Consider the simply supported beam with the associated influence line for the shear at point C in Fig. a. The maximum positive shear at point C is to be determined due to the series of concentrated (wheel) loads which move from right to left over the beam. The critical loading will occur when one of the loads is placed just to the right of point C, which is coincident with the positive peak of the influence line. By trial and error each of three possible cases can therefore be investigated, Fig. b. We have

$$\text{Case 1: } (V_C)_1 = 1(0.75) + 4(0.625) + 4(0.5) = 5.25 \text{ k}$$

$$\text{Case 2: } (V_C)_2 = 1(-0.125) + 4(0.75) + 4(0.625) = 5.375 \text{ k}$$

$$\text{Case 3: } (V_C)_3 = 1(0) + 4(-0.125) + 4(0.75) = 2.5 \text{ k}$$

Case 2, with the 1-k force located 5' ft from the left support, yields the largest value for V_C and therefore represents the critical loading. Actually investigation of Case 3 is unnecessary, since by inspection such an arrangement of loads would yield a value of $(V_C)_3$ that would be less than $(V_C)_2$.



$$\Delta V = Ps(x_2 - x_1)$$

Sloping Line

If the load moves past a point where there is a discontinuity or "jump" in the influence line, as point C in Fig. 6-27a, then the change in shear is simply

$$\Delta V = P(y_2 - y_1)$$

Jump (6-2)

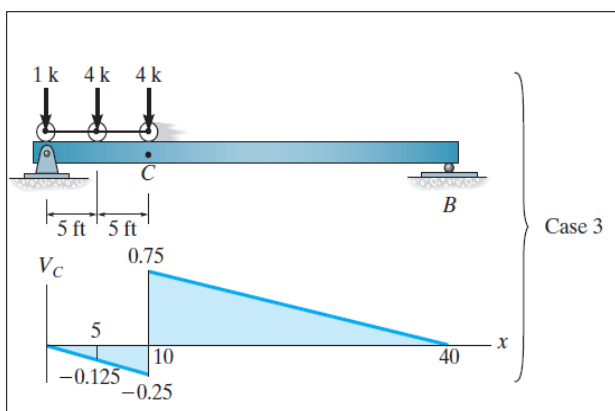
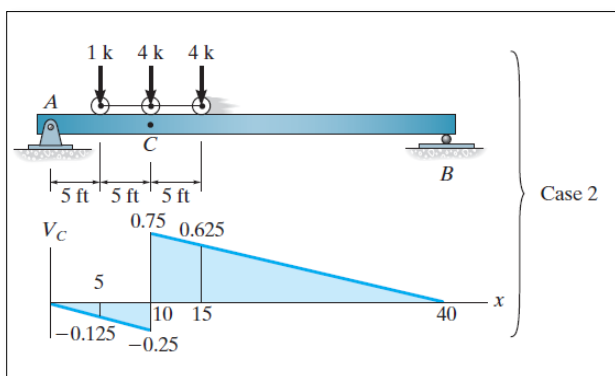
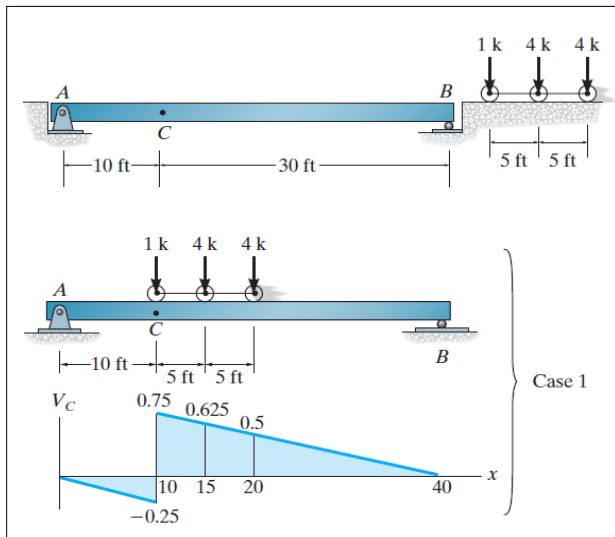
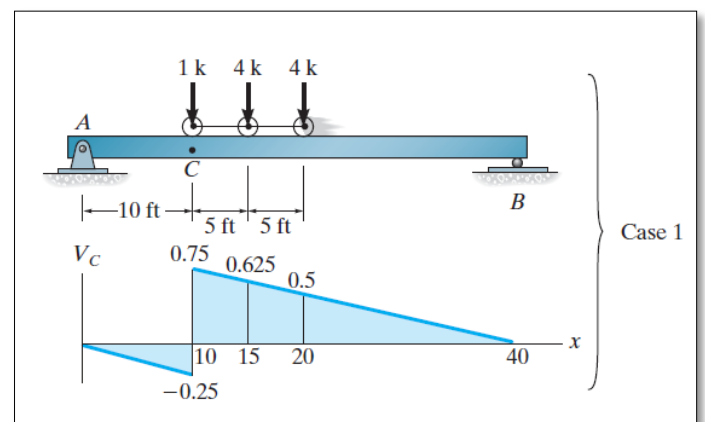
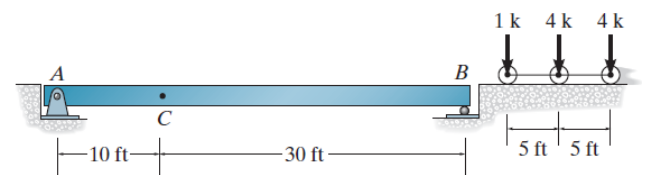
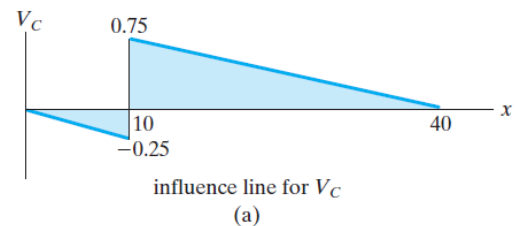
Use of the above equations will be illustrated with reference to the beam, loading, and influence line for V_C , shown in Fig. 6-28a. Notice that the magnitude of the slope of the influence line is $s = 0.75/(40 - 10) = 0.25/10 = 0.025$, and the jump at C has a magnitude of $0.75 + 0.25 = 1$. Consider the loads of Case 1 moving 5 ft to Case 2, Fig. 6-28b. When this occurs, the 1-k load jumps *down* (-1) and *all* the loads move *up* the slope of the influence line. This causes a change of shear,

$$\Delta V_{1-2} = 1(-1) + [1 + 4 + 4](0.025)(5) = +0.125 \text{ k}$$

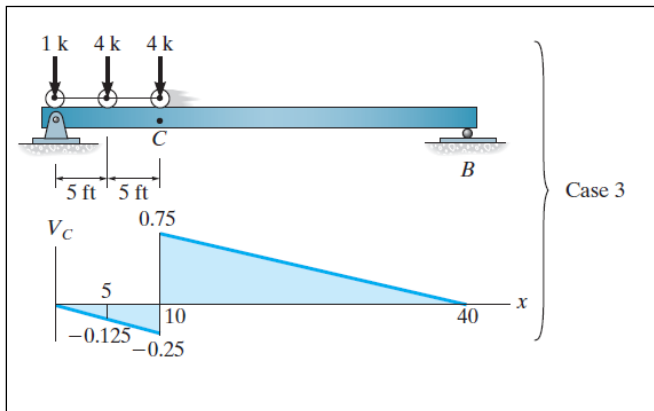
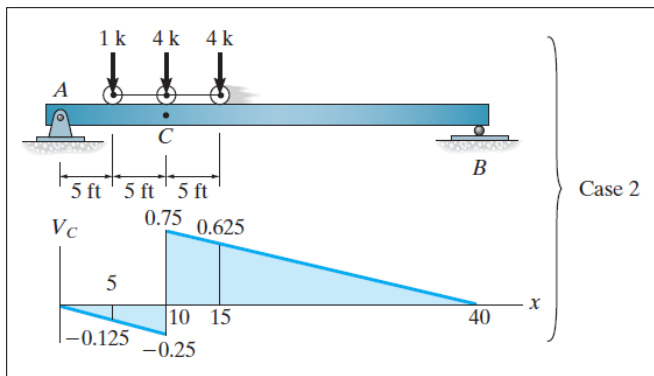
Since ΔV_{1-2} is positive, Case 2 will yield a larger value for V_C than Case 1. [Compare the answers for $(V_C)_1$ and $(V_C)_2$ previously computed, where indeed $(V_C)_2 = (V_C)_1 + 0.125$.] Investigating ΔV_{2-3} , which occurs when Case 2 moves to Case 3, Fig. 6-28b, we must account for the downward (negative) jump of the 4-k load and the 5-ft horizontal movement of all the loads *up* the slope of the influence line. We have

$$\Delta V_{2-3} = 4(-1) + (1 + 4 + 4)(0.025)(5) = -2.875 \text{ k}$$

Since ΔV_{2-3} is negative, Case 2 is the position of the critical loading, as determined previously.



The critical position of the loads can be determined in a more direct manner by finding the change in shear, ΔV , which occurs when the loads are moved from Case 1 to Case 2, then from Case 2 to Case 3, and so on. As long as each computed is *positive*, the new position will yield a larger shear in the beam at C than the previous position. Each movement is investigated until a negative change in shear is computed, ΔV . When this occurs, the previous position of the loads will give the critical value. The change in shear, ΔV , for a load P that moves from position x_1 to x_2 over a beam can be determined by multiplying P by the change in the ordinate of the influence line ($y_2 - y_1$), that is, If the slope of the influence line is s , then $(y_2 - y_1) = s(x_2 - x_1)$ and therefore



Moment. We can also use the foregoing methods to determine the critical position of a series of concentrated forces so that they create the largest internal moment at a specific point in a structure. Of course, it is first necessary to draw the influence line for the moment at the point and determine the slopes s of its line segments. For a horizontal movement $(x_2 - x_1)$ of a concentrated force P , the change in moment, ΔM , is equivalent to the magnitude of the force times the change in the influence-line ordinate under the load, that is,

$$\Delta M = Ps(x_2 - x_1) \quad (6-3)$$

Sloping Line

As an example, consider the beam, loading, and influence line for the moment at point C in Fig. 6-29a. If each of the three concentrated forces is placed on the beam, coincident with the peak of the influence line, we will obtain the greatest influence from each force. The three cases of loading are shown in Fig. 6-29b. When the loads of Case 1 are moved 4 ft to the left to Case 2, it is observed that the 2-k load decreases ΔM_{1-2} , since the slope $(7.5/10)$ is downward, Fig. 6-29a. Likewise, the 4-k and 3-k forces cause an increase of ΔM_{1-2} , since the slope $[7.5/(40 - 10)]$ is upward. We have

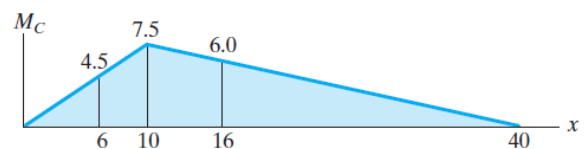
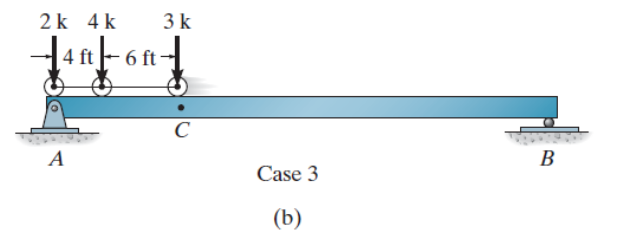
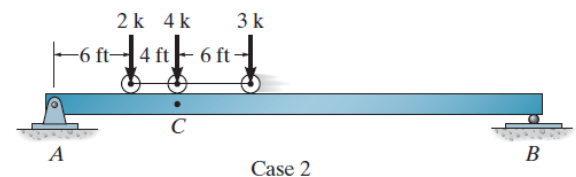
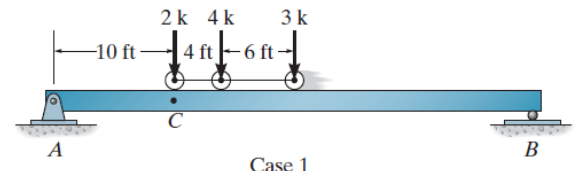
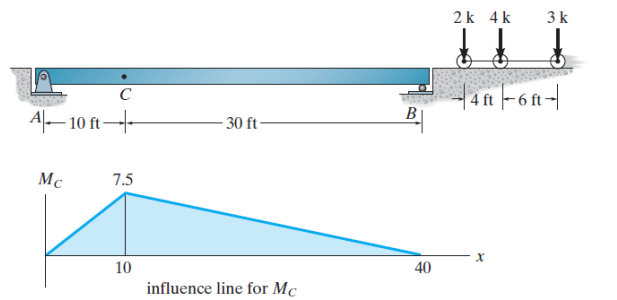
$$\Delta M_{1-2} = -2\left(\frac{7.5}{10}\right)(4) + (4 + 3)\left(\frac{7.5}{40 - 10}\right)(4) = 1.0 \text{ k} \cdot \text{ft}$$

Since ΔM_{1-2} is positive, we must further investigate moving the loads 6 ft from Case 2 to Case 3.

$$\Delta M_{2-3} = -(2 + 4)\left(\frac{7.5}{10}\right)(6) + 3\left(\frac{7.5}{40 - 10}\right)(6) = -22.5 \text{ k} \cdot \text{ft}$$

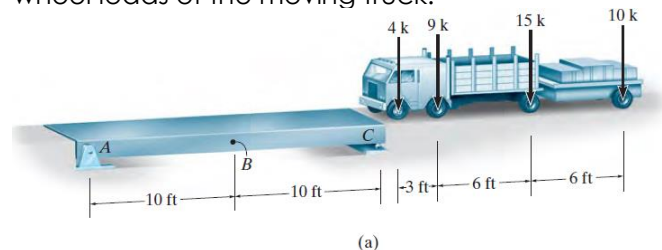
Here the change is negative, so the greatest moment at C will occur when the beam is loaded as shown in Case 2, Fig. 6-29c. The maximum moment at C is therefore

$$(M_C)_{\max} = 2(4.5) + 4(7.5) + 3(6.0) = 57.0 \text{ k} \cdot \text{ft}$$



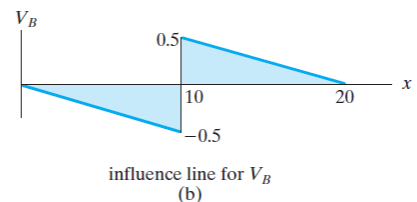
Example 1

Determine the maximum positive shear created at point B in the beam shown in Fig. a due to the wheel loads of the moving truck.



SOLUTION

The influence line for the shear at B is shown in Fig. 6-30b.



3-ft Movement of 4-k Load. Imagine that the 4-k load acts just to the right of point B so that we obtain its maximum positive influence. Since the beam segment BC is 10 ft long, the 10-k load is not as yet on the beam. When the truck moves 3 ft to the left, the 4-k load jumps *downward* on the influence line 1 unit and the 4-k, 9-k, and 15-k loads create a positive increase in ΔV_B , since the slope is upward to the left. Although the 10-k load also moves forward 3 ft, it is still not on the beam. Thus,

$$\Delta V_B = 4(-1) + (4 + 9 + 15)\left(\frac{0.5}{10}\right)3 = +0.2 \text{ k}$$

6-ft Movement of 9-k Load. When the 9-k load acts just to the right of B , and then the truck moves 6 ft to the left, we have

$$\Delta V_B = 9(-1) + (4 + 9 + 15)\left(\frac{0.5}{10}\right)(6) + 10\left(\frac{0.5}{10}\right)(4) = +1.4 \text{ k}$$

Note in the calculation that the 10-k load only moves 4 ft on the beam.

6-ft Movement of 15-k Load. If the 15-k load is positioned just to the right of B and then the truck moves 6 ft to the left, the 4-k load moves only 1 ft until it is off the beam, and likewise the 9-k load moves only 4 ft until it is off the beam. Hence,

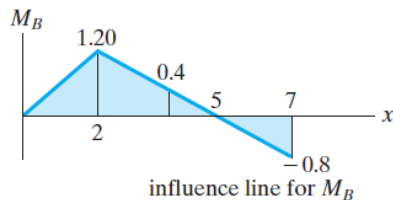
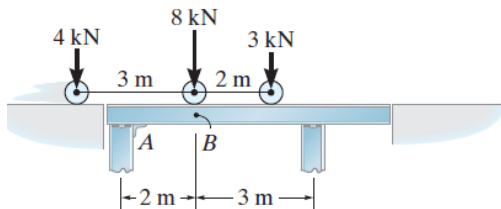
$$\Delta V_B = 15(-1) + 4\left(\frac{0.5}{10}\right)(1) + 9\left(\frac{0.5}{10}\right)(4) + (15 + 10)\left(\frac{0.5}{10}\right)(6) = -5.5 \text{ k}$$

Since ΔV_B is now negative, the correct position of the loads occurs when the 15-k load is just to the right of point B , Fig. 6-30c. Consequently,

$$(V_B)_{\max} = 4(-0.05) + 9(-0.2) + 15(0.5) + 10(0.2) = 7.5 \text{ k} \quad \text{Ans.}$$

Example 2

Determine the maximum positive moment created at point B in the beam shown in figure a due to the wheel loads of the crane.



SOLUTION

The influence line for the moment at B is shown in Fig. 6-31b.

2-m Movement of 3-kN Load. If the 3-kN load is assumed to act at B and then moves 2 m to the right, Fig. 6-31b, the change in moment is

$$\Delta M_B = -3\left(\frac{1.20}{3}\right)(2) + 8\left(\frac{1.20}{3}\right)(2) = 7.20 \text{ kN} \cdot \text{m}$$

Why is the 4-kN load not included in the calculations?

3-m Movement of 8-kN Load. If the 8-kN load is assumed to act at B and then moves 3 m to the right, the change in moment is

$$\begin{aligned} \Delta M_B &= -3\left(\frac{1.20}{3}\right)(3) - 8\left(\frac{1.20}{3}\right)(3) + 4\left(\frac{1.20}{2}\right)(2) \\ &= -8.40 \text{ kN} \cdot \text{m} \end{aligned}$$

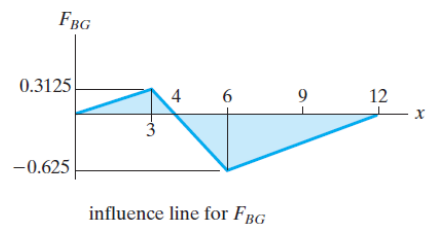
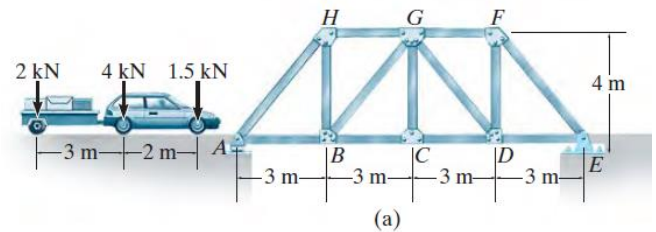
Notice here that the 4-kN load was initially 1 m off the beam, and therefore moves only 2 m on the beam.

Since there is a sign change in ΔM_B , the correct position of the loads for maximum positive moment at B occurs when the 8-kN force is at B , Fig. 6-31b. Therefore,

$$(M_B)_{\max} = 8(1.20) + 3(0.4) = 10.8 \text{ kN} \cdot \text{m} \quad \text{Ans.}$$

Example 3

Determine the maximum compressive force developed in member BG of the side truss in figure due to the right side wheel loads of the car and trailer. Assume the loads are applied directly to the truss and move only to the right.



SOLUTION

The influence line for the force in member BG is shown in Fig. 6-32b. Here a trial-and-error approach for the solution will be used. Since we want the greatest negative (compressive) force in BG , we begin as follows:

1.5-kN Load at Point C. In this case

$$\begin{aligned} F_{BG} &= 1.5 \text{ kN}(-0.625) + 4(0) + 2 \text{ kN}\left(\frac{0.3125}{3 \text{ m}}\right)(1 \text{ m}) \\ &= -0.729 \text{ kN} \end{aligned}$$

4-kN Load at Point C. By inspection this would seem a more reasonable case than the previous one.

$$\begin{aligned} F_{BG} &= 4 \text{ kN}(-0.625) + 1.5 \text{ kN}\left(\frac{-0.625}{6 \text{ m}}\right)(4 \text{ m}) + 2 \text{ kN}(0.3125) \\ &= -2.50 \text{ kN} \end{aligned}$$

2-kN Load at Point C. In this case all loads will create a compressive force in BC .

$$\begin{aligned} F_{BG} &= 2 \text{ kN}(-0.625) + 4 \text{ kN}\left(\frac{-0.625}{6 \text{ m}}\right)(3 \text{ m}) + 1.5 \text{ kN}\left(\frac{-0.625}{6 \text{ m}}\right)(1 \text{ m}) \\ &= -2.66 \text{ kN} \quad \text{Ans.} \end{aligned}$$

Since this final case results in the largest answer, the critical loading occurs when the 2-kN load is at C .